

Sky & Space: Class 9

A year of measurement and argument for class 9: coordinates, the evidence for heliocentrism, Kepler to gravitation to ISRO's orbits, the Sun as a star, a hand-plotted H-R diagram, the expanding universe — and the Indic tradition read with the same toolkit.

15 **36** **20 h** **4 h** **School: High (9–12)**
SESSIONS CONCEPTS TAUGHT TOTAL OUTDOORS AUDIENCE

Moon: start when the moon is **25.75–27.5 days old** (🌑 waning crescent → 🌑 waning crescent)

ASSUMED ALREADY KNOWN

The Pole Star Ecliptic & Zodiac Moon Orbits Earth Observing Logs & Sketching
 First Light: Using a Telescope Lunar Observing Dark Adaptation & Averted Vision
 Lenses, Mirrors & Focal Length Spectrum & Colour Waves The Sun is a Star Nebulae & Star Clusters
 Galaxies Gnomon & Local Noon The Solar System Family Hot & Cold: Temperature Models in Science
 The Navagrahas Gravity: Things Fall Newton's Third Law

Closed under prerequisites: 61 concepts in total are taken as prior knowledge.

1 The celestial sphere: a sky you can point at

CLASSROOM · 1 H 20 MIN

KIT Projector · Paper, pencils, protractor

L3 Celestial Sphere

A model sphere of infinite radius carrying the stars, with celestial poles and a celestial equator projected from Earth's own. It turns star positions into geometry we can compute.

L3 Altitude–Azimuth Coordinates

The local way to point: azimuth around the horizon from north, altitude up from the horizon. Simple and intuitive — but every star's alt-az changes by the minute.

TRAINER NOTE Open with the year's thesis: class 8 taught you to find things, class 9 teaches you to measure them, and measurement needs an address system. The celestial sphere must be sold honestly — it is FALSE (they know the stars are at wildly different distances from class 8) and USEFUL, which makes it the best model-thinking specimen in the course; say both sentences in the same breath. Then alt-az: altitude and azimuth of the classroom clock, the water tank, yesterday's moon. Drill the fatal flaw live — measure something's alt-az, wait forty minutes, measure again. The address expired. That failure is session 2's motivation, so do not fix it today.

2 Addresses that don't expire

INDOOR LAB · 1 H 20 MIN

KIT Projector · Star chart / planisphere · Paper, pencils, protractor

L3 Right Ascension & Declination

Sky coordinates that rotate with the stars: declination is celestial latitude, right ascension the sky's longitude measured in hours. A star's RA/Dec is (almost) permanent — this is how catalogues work.

L3 Sky from Different Latitudes

At the equator every star rises and sets vertically; at the poles nothing rises or sets at all. Your latitude completely decides which sky you own — Canopus for Chennai, none for London.

L3 Meridian & Transit

The meridian is the north–south line through your zenith; objects are highest — and best observed — when they cross (transit) it. Transit timing is the backbone of classical positional astronomy.

TRAINER NOTE Equatorial coordinates fix session 1's expiring address by bolting the grid to the turning sphere itself — latitude and longitude for the sky, with the celestial equator inherited from Earth's. The one diagram worth thirty minutes: pole star altitude = your latitude. Have them prove it with a drawing, not accept it — a Kanyakumari observer and a Srinagar observer see Dhruva at 8° and 34° and each can compute where the other stands. Meridian transit closes the session: every star crosses your north-south line once a night, at its highest, and crossing time = a clock. Plant the flag here: 'a wall that times transits is an instrument' — session 15 walks into Jantar Mantar over this exact bridge.

3 The case for the Sun: retrograde and the phases of Venus

CLASSROOM · 1 H 20 MIN

KIT Projector · Paper, pencils, protractor

L3 Retrograde Motion

Planets occasionally loop backward against the stars for weeks. It puzzled astronomers for two millennia — it's a simple overtaking effect between orbiting worlds.

L3 The Heliocentric Model

From Aristarchus and Aryabhata's rotating Earth to Copernicus and Galileo: putting the Sun at the centre explains retrograde loops and Venus's phases with one clean idea.

L3 Planetary Configurations

Opposition, conjunction, greatest elongation: the geometry words that say where a planet is relative to the Sun — and therefore when it rises and how bright it looks.

L3 Phases of Venus

Venus shows Moon-like phases tied to its size: a small full disc far away, a large thin crescent nearby. Galileo's telescope made this the killer evidence for heliocentrism.

TRAINER NOTE Pay class 8's promised debt: why do planets loop backwards? Run it as a trial, not an announcement — geocentrism was not stupid, it fit the naked-eye data for two thousand years. Retrograde first, as the awkward fact; then the overtaking-on-the-highway demonstration in the school ground (inner students walk faster; watch your neighbour drift backward against the far fence) — Mars's loop with zero epicycles. Configurations give the vocabulary (opposition, conjunction, elongation — why Venus never strays far from the Sun). Then the executioner: Galileo's Venus showing a FULL phase, impossible in Ptolemy's model where Venus is always sunward of us. Let a student explain why it is impossible; that is the argument won by evidence, not authority — the sentence the whole year is built on.

4 Kepler's laws, then Newton's why

CLASSROOM · 1 H 20 MIN

KIT Projector · Paper, pencils, protractor · Measuring tape

L4 Kepler's Laws

Orbits are ellipses with the Sun at a focus; planets sweep equal areas in equal times; period squared tracks distance cubed. Three rules that turned astronomy quantitative.

L4 Universal Gravitation

Newton showed one inverse-square force explains falling apples, the Moon's orbit and Kepler's laws together. Gravity is the engine of everything astronomy studies.

TRAINER NOTE Order matters more here than anywhere in the course: Kepler FIRST, as pure observed pattern — ellipses, equal areas, $T^2 \propto a^3$ — mined from Tycho's data with no idea why. Have them verify the third law themselves with a table of real planet periods and distances; the moment T^2/a^3 comes out constant eight times running is the moment they believe laws can hide in numbers. Draw ellipses with two pins and a loop of string, always. THEN Newton, as the answer to 'why these patterns?' — one inverse-square attraction explains all three at a stroke, plus the falling apple, plus the Moon. They are doing gravitation in physics class this very term; your job is the part physics class skips — that the law was worth finding because Kepler had found something needing explanation.

5 Orbits: falling forever, and India's rockets

CLASSROOM · 1 H 20 MIN

KIT Projector

L2 India in Space

Chandrayaan-3 landed at the lunar south pole, Mangalyaan orbited Mars on a first attempt, Aditya-L1 stares at the Sun, AstroSat scans the X-ray sky — and Gaganyaan comes next.

L3 Rockets & Launch Vehicles

A rocket throws mass backwards to be pushed forwards, and sheds stages as they empty so it never hauls dead tanks to orbit. India's ladder runs SLV-3 (Rohini, 1980, the first Indian orbital launch), ASLV, the PSLV workhorse, GSLV with its indigenous cryogenic upper stage, LVM3 which sent up Chandrayaan-3, and the small-satellite SSLV.

L4 Orbits & Artificial Satellites

An orbit is perpetual falling that keeps missing the ground. Speed and altitude set the orbit: ISS circles in 92 minutes, geostationary satellites hover over one longitude.

TRAINER NOTE Newton's cannonball kills the year's stubbornest misconception — that astronauts float because there is no gravity up there. Fire the cannon faster and faster until the fall misses the Earth forever; gravity up there is nearly full strength, and an orbit IS falling. Insist on the phrase 'falling around the Earth' until the class uses it unprompted. Then the payoff nobody expects to be local: ISRO flies on exactly this physics. Aryabhata to Chandrayaan-3's south-pole landing to Aditya-L1 staring at the Sun; PSLV as the reliable workhorse and why a rocket is Newton's third law with the reaction mass thrown downward. Launch eastward from Sriharikota and the Earth's spin is free velocity — let them work out why east. Careers land harder here than in any assembly talk.

6 The Sun through a filtered telescope

SOLAR SESSION · 1 H 20 MIN · DAY 45

SKY needs **Any moon** · The Sun

KIT Refractor · Solar filter (front aperture) · Tripod · Paper, pencils, protractor

L3 Sunspots & Solar Activity

The Sun's magnetic weather: spots, flares and mass ejections rising and falling on an 11-year cycle. It drives aurorae and can rattle satellites and power grids.

L3 Refractors & Reflectors

Refractors use a lens, reflectors a mirror, catadioptrics both. Each design trades cost, size, and image quality differently — there is no single 'best telescope'.

L3 Solar Observing (Safely)

NEVER look at the Sun without certified filters — projection and proper solar film show sunspots safely; hydrogen-alpha scopes reveal prominences. Safety protocol is the first lesson.

TRAINER NOTE Safety is the lesson, not the preamble: NOTHING about this session happens until every student has repeated the rule — no eye to any unfiltered optic pointed sunward, ever; a telescope is a magnifying glass and their retina has no pain nerves. Check the front-aperture filter for pinholes in front of the class, deliberately, so they see checking is normal. Teach the designs comparison on the hardware itself first (refractor bends, reflector bounces — open the dob's tube end and show the mirror) while pairs sketch. Then the Sun: sunspots as magnetic storms larger than Earth — draw today's spots, and if you can manage a second look days later, the rotation of the Sun falls out of two sketches. Log it like any observation; the Sun is the one star they will ever see as a disc, which is precisely what sessions 7–10 exploit.

7 How bright, how far: magnitudes and parallax

CLASSROOM · 1 H 20 MIN

KIT Projector · Paper, pencils, protractor

L3 The Inverse-Square Law

Double the distance, quarter the light — brightness falls as $1/d^2$. This single rule connects a star's true power, its distance, and how bright it looks to us.

L3 The Magnitude Scale

Astronomy's upside-down brightness scale: smaller numbers are brighter, and 5 magnitudes means $100\times$. Vega ≈ 0 , faintest naked-eye stars $\approx +6$, the Sun -26.7 .

L4 Stellar Parallax

Nearby stars shift minutely against the background as Earth orbits — the parallax that defines the parsec and anchors every distance in astronomy.

TRAINER NOTE The magnitude scale is historical debris — backwards (bigger = fainter) and logarithmic — so teach it as Hipparchus's fault and move on; five steps = $100\times$, Vega ≈ 0 , faintest naked eye ≈ 6 , done. Inverse-square is the physics under everything stellar: same torch, twice the distance, quarter the light — do it with a phone torch and a lux-meter app on the lab bench, four data points. Then the year's most elegant measurement: hold up a thumb, blink alternate eyes, watch it jump against the far wall. Baseline = eye spacing. Now let the baseline be the Earth's orbit and the thumb be a star: parallax, the ONLY rung of the distance ladder that is pure geometry, no assumptions. Be honest that the angles are fractions of an arcsecond — which is why the light year had to wait for telescopes, and why class 8's Alpha Centauri number was hard-won.

8 Starlight decoded: colour, temperature, spectra

CLASSROOM · 1 H 20 MIN

KIT Projector · Paper, pencils, protractor

L3 Star Colours & Temperature

A star's colour is a thermometer reading: red ~3,000 K, yellow ~6,000 K, blue-white 10,000 K+. Hotter objects glow bluer — physics you can check with your own eyes.

L4 Spectroscopy

Elements imprint fingerprint lines on a spectrum. From those lines alone we read a star's composition, temperature and motion — how we know what stars are made of.

TRAINER NOTE Start by breaking the childhood colour code: in a flame or a filament, BLUE is hotter than red — the iron-in-the-forge sequence runs dull red, orange, yellow, white, blue-white. So Betelgeuse (red) is cool and Rigel (blue) is fierce, and star colour is a thermometer read across light years. Then the deeper trick: a slit of light through a grating (a CD sliver taped to card works; build one per pair from stationery) shows lines — barcodes, one pattern per element, the same in the lab and in a star. Someone should gasp at the implication and if nobody does, say it plainly: we know what stars are MADE OF without going. Kill the 19th-century philosopher Comte's famous 'we shall never know the composition of stars' as the best wrong prediction in science. Both threads — temperature and identity — braid into session 9's diagram.

9 The H-R diagram, plotted by hand

INDOOR LAB · 1 H 20 MIN

KIT Paper, pencils, protractor · Projector

L4 Luminosity & Absolute Magnitude

Apparent brightness mixes true output with distance; absolute magnitude removes distance (all stars imagined at 32.6 ly). Only then can stars be honestly compared.

L4 The H-R Diagram

Plot luminosity against temperature and stars fall into families: the main sequence, giants, white dwarfs. One scatter plot that encodes how stars live and die.

TRAINER NOTE First close the brightness loophole: apparent magnitude confounds 'truly bright' with 'merely close', so park every star at the same standard distance (10 parsecs — via session 7's parallax and inverse-square) and compare honestly; that is absolute magnitude, and the Sun's humiliating drop from -27 to $+4.8$ is the number to dwell on. Then the year's centrepiece, and it must be done by HAND: give pairs a table of ~30 real stars (colour/temperature vs absolute magnitude), graph paper, forty minutes. Do not pre-draw the axes' meaning — let them discover the diagonal band themselves. The main sequence emerging out of their own plotted dots is the closest a classroom gets to the 1911 moment: stars are not scattered at random, and a scatter plot just became the map of stellar physics. Name the corners (giants, dwarfs) only after every pair has found the band.

10 The lives of stars

CLASSROOM · 1 H 20 MIN

KIT Projector

L4 Star Birth

Cold gas clouds collapse under gravity, heat up, and ignite fusion. The Orion Nebula is a stellar nursery in action, visible in binoculars.

L4 Lives of Stars

Stars balance gravity against fusion for millions to trillions of years, then swell to giants and die — gently as white dwarfs or violently as supernovae. Mass decides everything.

TRAINER NOTE Yesterday's map becomes today's story. Birth first: a cold cloud like class 8's Orion Nebula, gravity (session 4's, same law) pulling it together until the core lights — the main sequence band they plotted is not a road stars travel but where stars SIT while fusing hydrogen, the single most common misconception at this level; attack it explicitly with the 'photograph of a crowd' analogy — babies, adults, elders all in one frame, and the H-R diagram is that photograph. Mass is destiny: heavier stars burn brighter and die younger, profligately. Walk the Sun's own future — swelling to a red giant (the class 8 'Sun is a star' pivot now runs in reverse), shedding, ending as a white dwarf — with the honest timescale of five billion years, which is also the reassurance. End at the fact that makes teenagers sit up: the calcium in their bones was cooked inside a star that died before the Sun was born.

11 The expanding universe

CLASSROOM · 1 H 20 MIN

KIT Projector · Paper, pencils, protractor

L4 Doppler & Redshift

Motion stretches or squeezes light waves: approaching sources shift blue, receding ones red. Measured in spectral lines, it gives speeds of stars and galaxies to high precision.

L5 The Expanding Universe

Nearly every galaxy's light is redshifted, and farther means faster — space itself is stretching. Run the film backward and the universe had a beginning.

L5 Big Bang & Cosmic History

13.8 billion years from a hot dense start: relic microwave glow, primordial hydrogen and helium, and the growth of galaxies all tell one consistent story.

TRAINER NOTE Three ideas, one crescendo, and each stands on something the class already owns. Doppler: they have HEARD it (every passing bike horn drops in pitch); light does it too, and session 8's spectral barcodes are the ruler — lines shifted redward mean recession, and the shift is a speedometer. Hubble: nearly every galaxy's barcode is redshifted, and the farther, the faster. Run the balloon with sticker-galaxies and let every sticker be 'home' — everyone sees everyone else receding, no centre, which defuses 'where did it bang?' before anyone asks. Then run the film backwards to a hot dense beginning ~13.8 billion years ago. Be scrupulous about what Big Bang cosmology is — an extrapolation from measured expansion, model-thinking at full stretch — and end the physics year on the scale joke the course has earned: class 8 ended at Andromeda; class 9 ends at everything.

12 Observing night: star-hopping the first-quarter sky

OBSERVING NIGHT · 1 H 20 MIN · DAY 100

SKY needs **First quarter** · The Moon, Constellations, Bright stars, Double stars, Open clusters

KIT Binoculars · Dobsonian reflector · Eyepiece set · Star chart / planisphere · Red torch

L3 Star Hopping

Navigating from a bright star to a faint target through chart-matched patterns, hop by hop. The craft skill that turns a chart reader into an observer.

TRAINER NOTE One skill tonight, practised until it is theirs: navigating chart-to-sky in hops — anchor on a bright star, walk a pattern of steps each smaller than the binocular field, arrive at something invisible to the naked eye. Choose two hop routes for the season in advance and dry-run them yourself the night before; a teacher lost mid-hop teaches the wrong lesson. This is the year's coordinates made physical — the chart is in equatorial coordinates and their fist is still 10°. The first-quarter moon is deliberate, not a compromise: craters along the terminator are at their best through the dob for the queue, and the moon sets by midnight. Students who finish a hop early get the meridian question: which of tonight's targets is transiting right now, and how do you know? Log everything at the eyepiece.

13 Imaging night: the moon on their phones

IMAGING NIGHT · 1 H 20 MIN · DAY 101

SKY needs **First quarter** · The Moon

KIT Dobsonian reflector · Eyepiece set · Smartphone · Phone-to-eyepiece adapter · Red torch

L2 Smartphone Astrophotography

A phone held (or bracketed) at the eyepiece captures the Moon and planets; night modes catch constellations and even the Milky Way. The zero-cost gateway to imaging.

TRAINER NOTE The very next evening, same moon, one goal: every student leaves with a moon photograph **THEY** took, on **THEIR** phone. The terminator craters are the one astrophotography target that succeeds on the first attempt, which is why this night exists — success breeds astronomers, and a phone photo travels to the whole family by bedtime, which is your outreach programme running itself. Technique is three rules: adapter (or two-handed brace) against the eyepiece, tap-and-hold to lock focus and drag exposure **DOWN** — every phone will try to brighten the moon into a white blob — and let a timer or volume-button shutter fire the shot, never a screen jab. Have them shoot ten and keep one; culling is the real lesson. Print the best three for the classroom wall with the photographers' names.

14 Nakshatra, rashi, and the astrology question

CLASSROOM · 1 H 20 MIN

KIT Projector · Star chart / planisphere

L3 The 12 Rashis

The ecliptic cut into twelve 30° signs — Mesha through Meena — paralleling the Greek zodiac. In Indian usage the rashis stay fixed to the stars (sidereal), not the seasons.

L3 The 27 Nakshatras

The ecliptic divided into 27 lunar mansions of 13°20' each — one for each day of the Moon's sidereal circuit, each anchored to a marker star (yogatara).

L3 Astronomy vs Astrology

Same sky, different claims: astronomy predicts where planets are; astrology claims what they mean. One invites testing, the other resists it — a respectful, evidence-first discussion every Indian classroom deserves.

TRAINER NOTE Teach the Indic sky as what it technically is — coordinate systems centuries old. The 27 nakshatras divide the moon's monthly path (13°20' each, one per night of the sidereal month: the moon 'staying in a different lodge' nightly is an OBSERVATION their grandparents' calendar encodes); the 12 rashis divide the Sun's yearly path — the same ecliptic from class 8, sliced two ways for two clocks. Find the class's janma nakshatras on the chart; engagement is instant and personal. Then the question every class 9 room is silently asking, treated with respect and the year's own toolkit: astrology is a MODEL, so test it the way session 3 tested geocentrism — what does it predict, and do the predictions hold when checked blind? Present the evidence and let the model fail on its own terms. Never mock — half the room's families consult panchangs — but be exact: the failure of astrology as prediction and the brilliance of the astronomy underneath it are BOTH true, and the second is the inheritance.

15 Jantar Mantar: the observatory with no lens

INDOOR LAB · 1 H 20 MIN

KIT Gnomon / shadow stick · Measuring tape · Paper, pencils, protractor · Projector

L3 Traditional Instruments

Shanku (gnomon), ghati yantra (water clock), chakra yantra (graduated ring) and the astrolabe: pre-telescopic instruments that measured time and angle to surprising precision.

L3 Jantar Mantar Observatories

Jai Singh II's giant masonry instruments (1720s) — the Samrat Yantra sundial reads time to 2 seconds. Architecture-scale astronomy you can still walk through in Jaipur and Delhi.

TRAINER NOTE End the year in the school ground, closing the oldest loop in the course. Start from the humble gnomon they have used since class 6, and scale it in imagination: what if the stick were 27 metres tall? The shadow edge moves a measurable millimetre every few seconds — sheer SIZE buys precision, which is the entire engineering idea of Jai Singh's 1720s observatories. Tour the instruments as solved problems, not monuments: the Samrat Yantra is a sundial big enough to read to two seconds; the meridian instruments do exactly what session 2 promised a transit-timing wall could do. Then the closing exercise, pairs, on paper: design a naked-eye instrument to measure one thing — a star's altitude, local noon, the Sun's solstice extremes — using only masonry and geometry. Judge the designs against Jai Singh's. Some student's wall-with-a-slit will essentially reinvent one of his instruments, and that is the note to end the year on: the toolkit they now carry was always the whole game.

FULL KIT LIST

Binoculars	sessions 12
Dobsonian reflector	sessions 12, 13
Eyepiece set	sessions 12, 13
Gnomon / shadow stick <i>(consumable)</i>	sessions 15
Measuring tape	sessions 4, 15
Smartphone	sessions 13
Phone-to-eyepiece adapter	sessions 13
Projector	sessions 1, 2, 3, 4, 5, 7, 8, 9, 10, 11, 14, 15
Red torch	sessions 12, 13
Refractor	sessions 6
Solar filter (front aperture)	sessions 6
Star chart / planisphere	sessions 2, 12, 14
Paper, pencils, protractor <i>(consumable)</i>	sessions 1, 2, 3, 4, 6, 7, 8, 9, 11, 15
Tripod	sessions 6